Relative Loan Pricing as an Early-Warning Signal of Non-Performing Loans

Endogenous Screening under Borrower Opacity

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https://github.com/fgalarzac/ai-project

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1 Question, motivation, and track

A bank supervisor observes loan prices before portfolios mature. Can a bank's unusually high price serve as an early-warning signal for its future non-performing loan ratio (NPL ratio)? The sign is not obvious. A high price can worsen the applicant pool through adverse selection, but it also makes a performing loan more valuable and may induce the bank to screen more carefully.

Question. When does a higher loan spread relative to comparable banks predict a higher future NPL ratio, and when can it predict a lower one? This is a **Track B** project. My thesis proposed the two mechanisms in reduced form; this adaptation adds the bank's missing optimization problem, derives the screening response from primitives, and states the restrictions required for the predictive sign to reverse with borrower opacity. The claim is about forecasting, not a causal effect of forcing a bank to change its rate.

Index banks by j, comparable credit segments by s, pricing months by t, and forecast horizons by h. The empirical signal is built in two steps:

$$s_{jst} = i_{jst} - r_t, \qquad D_{jst} = s_{jst} - \text{median}_{k \neq j}\{s_{kst}\}, \tag{1}$$

where i_{jst} is the lending rate, r_t the policy rate, s_{jst} the spread over policy, and D_{jst} the spread relative to peer banks in the same segment and month. Thus $D_{jst} = 0.02$ means two percentage points above the peer median when rates are decimals. The observed outcome is

$$\text{NPL}_{js,t+h} = \frac{\text{non-performing loan balance}_{js,t+h}}{\text{total outstanding loan balance}_{js,t+h}}, \tag{2}$$

using the data provider's loan-classification rule.

2 Agent problem and economic mechanism

The model suppresses j, s, t to make the mechanism readable. Borrower opacity is $\kappa \in [0, 1]$: a larger value means applicants are harder to compare using readily available information. The timing is: (1) a segment with κ is given; (2) the bank posts the relative spread D; (3) price sorting forms an applicant pool; (4) the bank observes that pool and chooses screening effort e; and (5) repayment or default is realized.

Before screening, the applicant pool has default probability

$$q^0(D, \kappa) = \bar{q} + a(\kappa)D, \qquad \bar{q} \in (0, 1), \quad a(\kappa) > 0. \tag{3}$$

Symbol	**Meaning**
D	relative spread chosen in stage one
κ	borrower-segment opacity
$q^0,\ q$	default probability before and after screening
$e,\ \bar{e}$	screening effort and its upper bound
a	price-sorting sensitivity of the applicant pool
$g,\ c$	quality gain per effort unit and screening-cost shifter
$\bar{R}+D,\ F$	performing-loan payoff and effort-independent cost
L	potential loan volume

Table 1: Primitive objects in the model.

Here $\bar{q}$ is baseline default risk and $a(\kappa)$ is the sensitivity of pool quality to relative price. The positive sign captures the selection channel: a higher relative price loses more price-sensitive applicants and worsens the unscreened pool. Screening effort $e \in [0, \bar{e}]$ changes risk to

$$q(D, e, \kappa) = q^0(D, \kappa) - g(\kappa)e, \qquad g(\kappa) > 0, \tag{4}$$

where $g(\kappa)$ is the reduction in default probability produced by one unit of effort. Effort costs $c(\kappa)e^2/2$, with $c(\kappa) > 0$.

Let $R(D) = \bar{R} + D > 0$ be the payoff from a performing loan, F funding and operating cost independent of effort, and $L(D, \kappa) > 0$ potential loan volume, with $L_D < 0$. The bank solves

$$\max_{D \in \mathcal{D}} L(D, \kappa) \left\{ \max_{e \in [0, \bar{e}]} [1 - q^0(D, \kappa) + g(\kappa)e](\bar{R} + D) - F - \frac{c(\kappa)}{2} e^2 \right\}, \tag{5}$$

where the feasible price set $\mathcal{D} = [\underline{D}, \overline{D}]$ is compact. Continuity and compactness ensure a first-stage price solution. The topic result studies the continuation quality map $q^*(D, \kappa)$ for any observed relative price, including an equilibrium price or variation generated by bank-specific markup and cost shocks. A particular demand system would be needed for a closed-form D^*, but not for the conditional forecasting mechanism.

3 Expected first-order condition and result

Fix D. The marginal benefit of effort is the quality gain times the payoff from performance, $g(\kappa)(\bar{R}+D)$; its marginal cost is $c(\kappa)e$. The first- and second-order conditions are

$$g(\kappa)(\bar{R} + D) - c(\kappa)e = 0, \qquad \frac{\partial^2 \pi}{\partial e^2} = -c(\kappa) < 0. \tag{6}$$

For an interior solution,

$$e^*(D, \kappa) = \frac{g(\kappa)}{c(\kappa)}(\bar{R} + D). \tag{7}$$

A higher D raises the payoff from avoiding default and therefore raises screening effort. Substituting (7) into (4) gives

$$q^*(D, \kappa) = \bar{q} + a(\kappa)D - h(\kappa)(\bar{R} + D), \qquad h(\kappa) \equiv \frac{g(\kappa)^2}{c(\kappa)}. \tag{8}$$

The object h is *screening leverage*: the improvement in quality created by the optimal screening response to a marginally larger payoff. It is not the effort level itself. The central decomposition is

$$\boxed{\beta(\kappa) \equiv \frac{\partial q^*}{\partial D} = \underbrace{a(\kappa)}_{\text{worse selection}} - \underbrace{h(\kappa)}_{\text{stronger screening}}.} \tag{9}$$

Candidate Proposition 1 (opacity threshold). Suppose: (i) the solution is interior and $q^* \in (0,1)$; (ii) a and h are continuous; (iii) $a'(\kappa) < 0$; (iv) $h'(\kappa) > 0$; and (v) $a(0) > h(0)$ while $a(1) < h(1)$. Then a unique $\kappa^* \in (0,1)$ solves $a(\kappa^*) = h(\kappa^*)$, and

$$\beta(\kappa) \begin{cases} > 0, & \kappa < \kappa^*, \\ = 0, & \kappa = \kappa^*, \\ < 0, & \kappa > \kappa^*. \end{cases} \qquad \frac{\partial^2 q^*}{\partial D \partial \kappa} = a' - h' < 0. \tag{10}$$

Continuity and endpoint crossing give existence by the intermediate value theorem; $a' - h' < 0$ gives uniqueness. The key economic restriction is exposed by logarithmic differentiation:

$$\frac{h'}{h} = 2\frac{g'}{g} - \frac{c'}{c}. \tag{11}$$

Therefore, higher screening cost alone cannot create the negative slope: it reduces h. Opacity must raise the marginal information gain g fast enough to dominate any rise in c. This is a falsifiable sufficient condition, not an implication built into the definition of opacity.

The theory's q^*_{jst} is default probability for newly originated credit, whereas $\text{NPL}_{js,t+h}$ is a later stock ratio. I connect them through the explicit forecasting map

$$\text{NPL}_{js,t+h} = \alpha_h + \rho_h q^*_{jst} + u_{js,t+h}, \qquad \rho_h > 0. \tag{12}$$

If this monotone map is supported, the predictive slope has the sign of $\rho_h \beta(\kappa)$, hence the sign of β. Equation (12) is a bridge to measurement, not a causal structural equation.

4 Why this is not already solved

The ingredients exist separately. Varian (1980) explains price dispersion using informed and uninformed buyers, but has neither screening nor default. Broecker (1990) combines rate competition with imperfect credit tests, but does not make testing a continuous effort choice. Ruckes (2004) endogenizes screening intensity as economic prospects change, but does not characterize the sign of a relative-price signal across opacity segments. Dell'Ariccia and Marquez (2006) shows how information asymmetry across banks selects screening or pooling regimes, rather than deriving a continuous $\partial q^*/\partial D$ decomposition. The narrow contribution here is to combine relative-price sorting with endogenous screening and identify the condition for a unique opacity threshold.

5 Plan, validation, and risks

I will (1) write a full proof of Proposition 1 and a robustness result for a general increasing payoff $R(D)$; (2) simulate the complete admissible parameter region instead of reporting only one calibration; (3) test whether D improves out-of-sample NPL-ratio forecasts by opacity group and horizon; and (4) formalize the threshold theorem in Lean after its assumptions are frozen. Current code symbolically checks the FOC, SOC, and derivative and numerically checks crossing, monotonicity, interior effort, and probability bounds.

The main risk is that $h'(\kappa) > 0$ fails. I will then report the three regions $a > h$, $a = h$, and $a < h$ without forcing a threshold. A second risk is endogenous pricing. The empirical fallback is explicitly predictive: bank and segment-by-month controls, rolling out-of-sample evaluation, and no causal interpretation of the price coefficient.

References

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